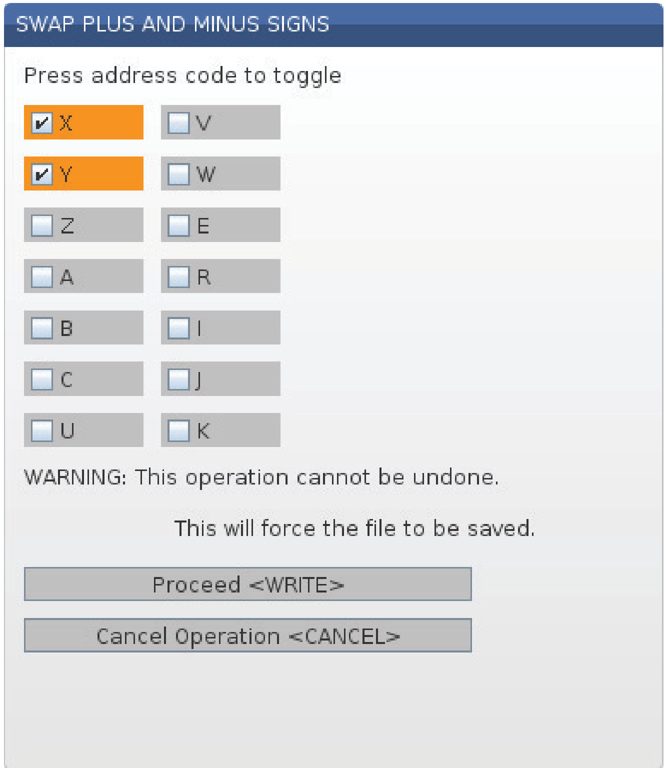
F5.5: Reverse Plus and Minus Signs Menu



- **`Reverse X And Y`**: Changes the X address codes in the program to Y address codes, and changes Y address codes to X address codes.

5.3 Tips and Tricks

The following sections provide insight into efficiently programming your Haas Turning Center.

5.3.1 Tips and Tricks - Programming

Short programs looped many times do not reset the chip conveyor if the intermittent feature is activated. The conveyor continues to start and stop at the commanded times. Refer to page **388** for information on the conveyor interval settings.

The screen displays the spindle and axis loads, the current feed and speed, positions, and the currently active codes while a program runs. Different display modes change the information that is displayed.

To clear all offsets and macro variables, press **[ORIGIN]** at the **`Active Work Offset`** screen. The control displays a popup menu. Pick **`Clear Work Offsets`** for the displayed message *`Are you sure you want to Zero(Y/N)`*. If `Y` is entered, all the work offsets (macros) in the area being displayed are set to zero. The values in the **`Current Commands`** display pages can be cleared as well. The Tool Life, Tool Load, and Timer registers are cleared by selecting the one to clear and pressing **[ORIGIN]**. To clear everything in a column, scroll to the top of the column onto the title and press **[ORIGIN]**.

To select another program enter the program number (`Onnnnn`) and press the arrow up or down. The machine must be in either **`Memory`** or **`Edit`** mode. To search for a specific command in a program, use Memory or Edit mode. Enter the address code (`A`, `B`, `C` etc.), or the address code and the value (`A1.23`), and press the up or down arrow key. If the address code is entered without a value, the search stops at the next use of that letter.

To transfer or save a program in MDI to the list of programs position the cursor at the beginning of the MDI program, enter a program number (`Onnnnn`), and press **[ALTER]**.

Program Review - Program Review allows the operator to move the cursor and review a copy of the active program on the right side of the display screen, and view the same program as it runs on the left side of the screen. To display a copy of the active program in the **`Inactive Program`** display, press **[F4]** while the **`Edit`** pane contains the active program.

Background Edit - This feature edits while a program runs. Press **[EDIT]** until the background **`Edit`** pane (on the right side of the screen) is active. Select a program to edit from the list and press **[ENTER]**. Press **[SELECT PROGRAM]** from this pane to select another program. Edits are possible as the program runs, however, edits to the running program will not take effect until the program ends with an `M30` or **[RESET]**.

Graphics Zoom Window - **[F2]** activates the zoom window when in **`Graphics`** mode. **[PAGE DOWN]** zooms in and page up expands the view. Use the arrow keys to move the window over the desired area of the part and press **[ENTER]**. Press **[F2]** and **[HOME]** to see full table view.

To Copy Programs - In**`Edit`** mode, a program can be copied into another program, a line, or a block of lines in a program. Define a block with the **[F2]** key, then move the cursor to the last program line to define, press **[F2]** or **[ENTER]** to highlight the block. Select another program to copy the selection to. Move the cursor to the point where the copied block is moved and press **[INSERT]**.

To Load Files - Select multiple files in the device manager, then press **[F2]** to select a destination.

To Edit Programs - Press **[F4]** while in **`Edit`** mode to display another version of the current program in the right-hand pane. Different portions of the programs can be edited alternately by pressing **[EDIT]** to switch from one side to the other. The program is updated once switched to the other program.

To Duplicate a Program - An existing program can be duplicated in List Program mode. To do this, select the program number to duplicate, type in a new program number (`Onnnnn`) and press **[F2]**. This can also be done through the popup help menu. Press **[F1]**, then select the option from the list. Type the new program name and press **[ENTER]**.

Several programs can be sent to the serial port. Highlight the desired programs from the program list to select them and press **[ENTER]**. Press **[SEND]** to transfer the files.

5.3.2 Offsets

To enter offsets:

1. Press **[OFFSET]** to toggle between the **`Tool Geometry`** and **`Work Zero Offset`** panes.
2. Press **[ENTER]** to add the entered number to the cursor-selected value.
3. Press **[F1]** to overwrite the cursor selected offset register with the entered number.
4. Press **[F2]** to enter the negative value into the offset.

5.3.3 Settings

The **[HANDLE JOG]** control is used to scroll through settings and tabs, when not in jog mode. Enter a known setting number and press the up or down arrow key to jump to the entered setting.

The Haas control can power off the machine using settings. These settings are: Setting 1 turns off power after machine is idle for `nn` minutes, and Setting 2 turns off power when `M30` is executed.

Memory Lock (Setting 8) when On, memory edit functions are locked out. When Off, memory can be modified.

Dimensioning (Setting 9) changes from **`Inch`** to **`MM`**. This changes all offset values too.

Reset Program Pointer (Setting 31) turns on and off the program pointer returning to the program beginning.

Scale Integer F (Setting 77) changes the interpretation of a feed rate. A feed rate can be misinterpreted if there is not a decimal point in the `Fnn` command. The selections for this setting are **`Default`**, to recognize a 4 place decimal. Another selection is **`Integer`** which recognizes a feed rate for a selected decimal position, for a feed rate that does not have a decimal.

Max Corner Rounding (Setting 85) is used to set the corner rounding accuracy required by the user. Any feed rate up to the maximum can be programmed without the errors getting above that setting. The control slows at corners only when needed.

Reset Resets Override (Setting 88) turns on and off the Reset key setting the overrides back to 100%.

Cycle Start/Feed hold (Setting 103) when `On`, **[CYCLE START]** must be pressed and held to run a program. Releasing **[CYCLE START]** generates a Feed Hold condition.

Jog Handle to Single Block (Setting 104) allows the **[HANDLE JOG]** control to be used to step through a program. Reverse the **[HANDLE JOG]** control to generate a Feed Hold condition.

Offset Lock (Setting 119) prevents the operator from altering any of the offsets.

Macro Variable Lock (Setting 120) prevents the operator from altering any of the macro variables.

5.3.4 Operation

[MEMORY LOCK] key switch - prevents the operator from editing programs and from altering settings when in the locked position.

[HOME G28] - Returns all axes to machine zero. To send just one axis to machine home, enter the axis letter and press **[HOME G28]**. To zero out all axes on the `Distance-To-Go` display, while in `Jog` mode, press any other operation mode (**[EDIT]**, **[MEMORY]**, **[MDI/DNC]**, etc.) then press **[HANDLE JOG]**. Each axis can be zeroed independently to show a position relative to the selected zero. To do this go to the `Position Operator` page, press **[HANDLE JOG]**, position the axes to the desired position and press **[ORIGIN]** to zero that display. In addition a number can be entered for the axis position display. To do this, enter an axis and number, for example, `X2.125` then **[ORIGIN]**.

`Tool Life` - Within the `Current Commands` page there is a `Tool Life` window displaying tool usage. This register counts each time the tool is used. The tool life monitor stops the machine once the tool reaches the value in the alarms column.

`Tool Overload` - Tool load can be defined by the Tool Load monitor; this changes normal machine operation if it reaches the tool load defined for that tool. When a tool overload condition is encountered, one of four actions occurs depending on Setting 84:

- `Alarm` - Generate an alarm
- `Feedhold` - Stop the feed
- `Beep` - Sounds an audible alarm
- `Autofeed` - Automatically increase or decrease the feed rate

Spindle speed is verified by checking the `Current Commands All Active Codes` display (also displayed in the Main Spindle window). Live tooling spindle axis RPM is also displayed on this page.

To select an axis for jogging, enter the axis name on the input line and press **[HANDLE JOG]**.

The Help display has all the G and M codes listed. They are available within the first tab of the Help tabbed menu.